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WalkNet: A Vision-Based Framework for Mapping Urban Pedestrian Infrastructure

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Abstract

Pedestrian infrastructure is inadequately or inconsistently mapped in most cities, limiting efforts to assess walkability, accessibility, and equitable transportation systems. This study introduces **WalkNet**, a city-wide scalable framework that generates detailed, georeferenced pedestrian networks by integrating street-level imagery with open geospatial data. The framework applies a semantic segmentation model on panoramic images to extract sidewalks and roads, then computes a sidewalk-to-road height ratio as a geometric proxy for road type. It classifies roads as arterial, collector, or residential using the ratio, aligning with U.S. transportation codes. We implemented WalkNet in Phoenix, Arizona, a city with a diverse urban form. Using the Segment Anything model, Google Street View imagery, and OpenStreetMap, WalkNet achieved **78.2%** overall road classification accuracy, **98.05%** and **78.19%** for arterial and residential roads, respectively. Collector roads showed high recall (**96.43%**) but lower precision due to overlap with residential features. A manual evaluation confirmed that the representation of crosswalks, park trails, and informal footways in OpenStreetMap often does not match real-world conditions, highlighting key gaps in existing open pedestrian maps. The results show that visually derived metrics can support city-scale infrastructure mapping. Researchers and city planners can use the resulting data to support thermal comfort routing, walkability analyses, and climate-responsive urban planning. In addition, Walknet is a low-cost alternative to comprehensively map pedestrian infrastructure compared to classical field-surveying methods.

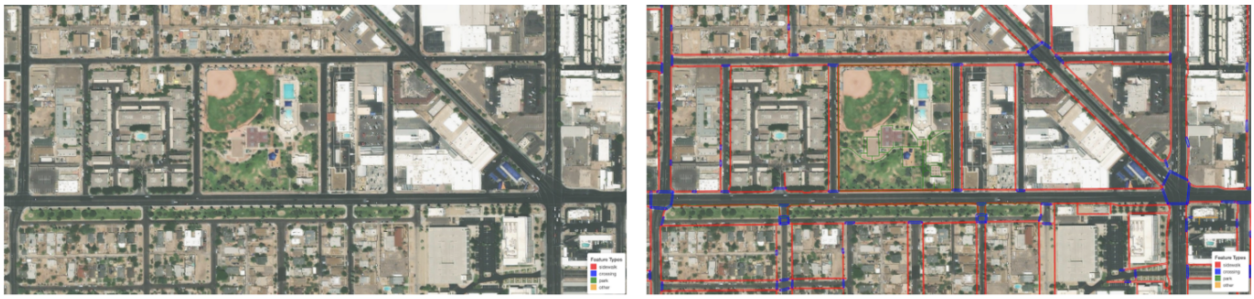


Figure 1: In the figure, there is an example of Pedestrian Network Mapping. On the left, a sample area in Phoenix, Arizona, before processing. On the right, the same area after applying WalkNet, with different walkable features color-coded for clarity. Sidewalks are shown in red, crosswalks in blue, parks in green, and other walkable features in yellow.

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